

HW/Lab 6: MIMO Implementation

ECSE 4560/6560: Modern Communication Systems

RPI Fall 2025

Goals and Instructions

The goals for this assignment are:

- Implement multiple forms of MIMO diversity techniques and understand their performance in terms of BER-SNR curves.
- Understand OFDM-MIMO system implementation using practical WiFi-based waveforms.

Follow these instructions:

- **Collaboration:** You can discuss concepts, but write your own solutions and code. **Any plagiarism may result in 0 points.**
- The assignment has one programming assignment.
- **Tools:** We will primarily use MATLAB. Directions use standard MATLAB functions. *Strongly recommended* toolboxes: Signal Processing, Communications. (Base-MATLAB fallbacks are acceptable.)
- **Submission:** Upload a single PDF with your answers and plots (not including code) to Gradescope before the deadline. Also, compress (into a ZIP file) and upload your finished ‘lab6 starter code’ folder to a separate Gradescope portal before the same deadline. Please do not use MATLAB publish for this homework.
- Total 20 points for ECSE 4560 and 25 points for ECSE 6560 students.

1 OFDM-MIMO Decoding

Consider an OFDM transmitter sending bit sequence b , which could be 0 or 1 with equal probability. The transmitter uses BPSK modulation, i.e., it sends -1 if b is 0, and $+1$ if b is 1. The symbols are loaded into 64 subcarriers with an appropriate guard band and are converted into time samples using IFFT, and an appropriate CP is added. Assume that the channel introduces additive Gaussian random noise with zero mean, and a standard deviation of σ besides channel response. We also use a legacy OFDM preamble (STS and 2.5 copy of LTS) along with MIMO preamble (One LTS each per antenna per time-slot for MIMO channel estimation).

1. **Case 1: (1x2 MIMO)** Consider a case where the transmitter has one antenna, and the receiver has two antennas, with the received symbol as $y_{i,A} = h_A x_i + n_{i,A}$ and $y_{i,B} = h_B x_i + n_{i,B}$, where the $n_{i,A}$ and $n_{i,B}$ are i.i.d. random Gaussian variables, with zero mean and standard deviation of σ . The index i represents time-domain samples. What is the receiver's decoding rule that gives the best estimate of b_i , given $y_{i,A}$, $y_{i,B}$, h_A , h_B , and σ ? Also, provide the analytical expression for SNR.
Hint: Use maximum ratio combining.
2. **Case 2: (1x4 MIMO)** Consider a case where the transmitter has one antenna, and the receiver has four antennas with the received symbol as $y_{i,A} = h_A x_i + n_{i,A}$, $y_{i,B} = h_B x_i + n_{i,B}$, $y_{i,C} = h_C x_i + n_{i,C}$, $y_{i,D} = h_D x_i + n_{i,D}$ where $n_{i,A}, n_{i,B}, n_{i,C}, n_{i,D}$ are i.i.d. random Gaussian variables, with zero mean and standard deviation of σ . What is the receiver's decoding rule that gives the best estimate of b_i , given $y_{i,A}, y_{i,B}, y_{i,C}, y_{i,D}, h_A, h_B, h_C, h_D$ and σ ? Provide the analytical expression for SNR and compare it with the previous case 1, which one is better?
3. **Case 3: (2x1 MIMO)** Consider a case where transmitter has two antenna and is only transmitting one stream like signal, and the receiver has one antenna, with the received symbol as $y_i = h_A x_{i,A} + h_B x_{i,B} + n_i$, where n_i is a random Gaussian variable, with zero mean and standard deviation of σ . The transmit precoding implements $x_{i,A} = c_{i,A} x_i$ and $x_{i,B} = c_{i,B} x_i$, where x_i is a single stream of data (indexed by time sample i), and $c_{i,A}, c_{i,B}$ are precoding coefficients. What is the transmitter's precoding / receiver's decoding rule that gives the best estimate of b_i , given y_i, h_A, h_B , and σ ? Provide the analytical expression for SNR and compare it with the expressions from Case 1 and Case 2; which one is better?

We will implement these examples for the OFDM-MIMO system using the given starter code.

Starter code

- You are given the starter code 'lab6_starter_code.zip' on Piazza. The main script is 'hw6.m'. The default parameter settings in 'hw6.m' are as follows.

```

1 SNR_list = 0:1:10; % SNR in dB
2 N_iter = 100; % number of Monte-Carlo iterations
3
4 USE_CHANNEL_ESTIMATES = 0;
5 CFO_FLAG = 0; % flag to enable CFO
6 DETECTION_OFFSET = 100; % to add packet detection error
7 LTS_CORR_THRESH = .8; % correlation threshold for LTS detection
8 DO_APPLY_CFO_CORRECTION = 0; % flag to enable CFO correction

```

The remaining parameters are based on OFDM processing as seen in HW4.

- Running 'hw6.m' will plot the simulated BER values for the three MIMO cases. Since we set small SNR values, packet detection failure or error could happen. Packet detection failure will cause a warning 'No LTS Correlation Peaks Found!', and the code will automatically skip this Monte-Carlo iteration, so we can ignore it. But packet detection error will cause an error warning 'something wrong' and stop the simulation. In this case just rerun the simulation.
- For debugging or testing, you can save some time by setting a single SNR value and a smaller number of iterations, and 'hw6.m' will generate a plot with single dot for each case as shown below:

```

1 SNR_list = 10; % SNR in dB
2 N_iter = 10; % number of Monte-Carlo iterations

```

- For deliverables, please set them **back to default**.

```

1 SNR_list = 0:1:10; % SNR in dB
2 N_iter = 100; % number of Monte-Carlo iterations

```

What you need to implement

1. Replace the P-code function 'tx_precoding.p' by your own code 'tx_precoding.m', so 'hw6.m' can still run correctly. This function is directly called in 'hw6.m' to perform precoding for **Case 3 (MIMO 2x1)**.
2. Replace the P-code function 'mimo_processing.p' by your own code 'mimo_processing.m', so 'hw6.m' can still run correctly. This function is called in function 'MIMO_OFDM_RX.m' to perform signal combining for the three cases.
 - Input 'syms_f_mat_1A' is a matrix, where each column is for an OFDM symbol.
 - Input 'h1A' is a constant ground truth channel value, which is defined in 'hw6.m'.
 - Output 'syms_eq_mat_pilots_case1' has the identical size to 'syms_f_mat_1A', which captures the MIMO receiver combining for case 1. Similarly, define for cases 2 and 3.
 - Note that we have done precoding for case 3 in the previous step.
3. (**ECSE 6560 only**) Now, in 'hw6.m', turn ON the flag for channel estimate as below.


```

1 USE_CHANNEL_ESTIMATES = 1;

```

This will enable MIMO processing to be done at the receiver based on *channel estimates* (already provided to you) rather than assuming the *ground truth channel* at the receiver. Then, in the function 'MIMO_OFDM_RX.m', we can see that the inputs of the function mimo_processing() change. Modify your 'mimo_processing.m' code so 'hw6.m' can work correctly in this setting.

Deliverables

1. Answer the above three questions for the three cases. Submit your answers.
2. Run 'hw6.m' to generate a BER plot for the three cases using the settings below. Are the BER values as expected? Explain for the same BER target, which case requires a higher SNR? Explain your answer. Submit the plot with your explanation.

```

SNR_list = 0:1:10;
N_iter = 100;
USE_CHANNEL_ESTIMATES = 0;

```

3. (**ECSE 6560 only**) Redo the previous part (deliverable 2) with the following setting. Submit the BER-SNR plot with an explanation for changes in the plot compared to deliverable 2.

```

SNR_list = 0:1:10;
N_iter = 100;
USE_CHANNEL_ESTIMATES = 1;

```

4. (ECSE 6560 only) In the code, we did not use the channel estimates ('rx_H_est_1A',...) in the MRC precoding for **Case 3 (2x1 MIMO)**. How would you modify the code so we can use channel estimates ('rx_H_est_1A',...) for MRC precoding? More specifically, how to obtain the channel estimates ('rx_H_est_1A',...), and how to perform precoding with the channel estimates ('rx_H_est_1A',...)? Explain briefly and submit your answer. (You are not required to implement the code for this part)
5. We do not ask for CFO correction in this assignment, however, you can play with the code if you like.
6. Upload a single PDF with your answers and plot (no need to include code) to Gradescope. Also, compress (into a ZIP file) and upload your finished 'lab6 starter code' folder to a separate Gradescope portal.